

Workspace-Bench 1.0: Benchmarking AI Agents on Workspace Tasks with Large-Scale File Dependencies

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Research Questions

- How can we systematically evaluate AI agents' ability to handle realistic workspace tasks with complex file dependencies?
- What are the key capability bottlenecks preventing current AI agents from performing well on workspace learning tasks?
- How do different agent harnesses and foundation models perform on tasks requiring cross-file reasoning and dependency understanding?

Core Idea

Current AI agent benchmarks evaluate agents on isolated files or simplified scenarios, missing the complex reality of workplace tasks where agents must navigate thousands of interconnected files with implicit dependencies. The key insight is that workspace learning requires understanding file relationships, version lineages, and semantic connections across heterogeneous formats - capabilities that existing benchmarks don't test. Workspace-Bench addresses this by creating realistic workspaces with 20,476 files across 5 professional personas, forcing agents to discover dependencies rather than being spoon-fed relevant files.

Key Formula

Node F1 Score

$$NF1 = \frac{2 \cdot NP \cdot NR}{NP + NR}$$

Measures how well agents identify relevant files in dependency graphs. NP is node precision (predicted nodes that are correct), NR is node recall (ground truth nodes that are found). Critical for evaluating workspace exploration capability.

Edge F1 Score

$$EF1 = \frac{2 \cdot EP \cdot ER}{EP + ER}$$

Measures how well agents understand file dependencies and relationships. EP is edge precision, ER is edge recall comparing predicted vs ground truth dependency edges. Key metric for lineage tracing evaluation.

Key Figure



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Abstract

Workspace learning requires AI agents to identify, reason over, exploit, and update explicit and implicit dependencies among heterogeneous files in a worker’s workspace, enabling them to complete both routine and advanced tasks effectively. Despite its importance, existing relevant benchmarks largely evaluate agents on pre-specified or synthesized files with limited real-world dependencies, leaving workspace-level evaluation underexplored. To this end, we introduce **Workspace-Bench**, a benchmark for evaluating AI agents on **Workspace Learning** involving Large-Scale File Dependencies. We construct realistic workspaces with 5 worker profiles, 74 file types, 20,476 files (up to 20GB) and curate 388 tasks, each with its own file dependency graph, evaluated across 7,399 total rubrics that require cross-file retrieval, contextual reasoning, and adaptive decision-making. We further provide **Workspace-Bench-Lite**, a 100-task subset that preserves the benchmark distribution while reducing evaluation costs by about 70%. We evaluate 4 popular agent harnesses and 7 foundation models. Experimental results show that current agents remain far from reliable workspace learning, where the best reaches only 68.7%, substantially below the human result of 80.7%, and the average performance across agents is only 47.4%.

Project Page: <https://github.com/OpenDataBox/Workspace-Bench>

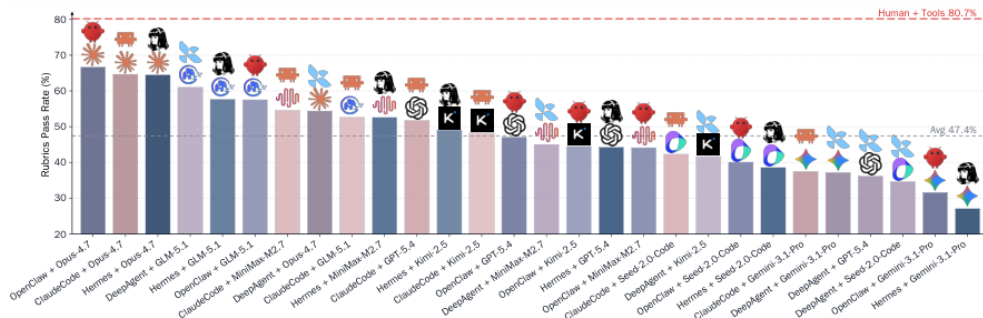


Figure 1 Performance on the lite version of **Workspace-Bench**. We evaluate four harnesses (ClaudeCode, DeepAgent, Hermes, and OpenClaw) with seven backbone LLMs. The top three configurations all pair Opus-4.7 with different harnesses, while lower-ranked combinations exhibit substantial variation across both harnesses and LLMs.

Figure 1: Shows performance of 28 agent configurations on Workspace-Bench-Lite, revealing that even the best agents achieve only 68.7% vs 80.7% human performance, with most configurations clustered around 47.4% average.

Limitations

- Limited to 5 professional personas, may not cover all workplace scenarios
- Human annotation process is expensive and time-consuming (>2500 hours)
- Evaluation relies on Agent-as-a-Judge paradigm which may have biases
- Workspace-Bench-Lite subset reduces evaluation cost but covers only 100/388 tasks